HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed **Python** backend services for energy data processing, integrating with PostgreSQL databases to support national energy modelling frameworks.
- Built and maintained **PostgreSQL** relational databases for storing large-scale energy consumption and projection data, ensuring data integrity and efficient querying.
- Developed and enhanced **data pipelines** using Python for ETL processes, moving data between various storage solutions and analytical tools, supporting reliable energy projections.
- Contributed to the development of applications used to visualise and compare **energy modelling** results, translating complex data into actionable insights for stakeholders.
- Implemented robust software testing strategies, including unit and integration tests, to ensure the quality and reliability of production-level energy modelling software.
- Managed code versions and maintained comprehensive documentation for energy modelling tools and systems, adhering to high standards of software engineering best practices.
- Containerized backend services using **Docker** and deployed them to **AWS** EC2 instances, managing configurations and ensuring smooth integration with cloud-based storage solutions.

SKILLS

Languages – Python, SQL, R

Databases – PostgreSQL, MySQL, Oracle

Cloud – AWS (EC2, S3, Lambda, CloudWatch)

Tools – Git, GitHub, Docker, Jenkins, Jira, Confluence, Splunk, Dynatrace

Frameworks/Libraries – Spring Boot, pandas, NumPy, scikit-learn

Analytics/BI – Power BI, Tableau, DAX, Power Query, data modelling, data warehousing

Domain – Energy Modelling, Data Pipelines, Backend Development, Software Testing, Version Control

PROJECTS

Energy Data Processing Pipeline (HCL Technologies)

- Engineered a **Python**-based data pipeline to ingest and process terabytes of historical energy consumption data from disparate sources into a centralized **PostgreSQL** data warehouse.
- Developed automated ETL jobs using pandas and SQL scripts to clean, transform, and validate energy datasets, reducing manual data preparation effort by approximately **40%**.
- Implemented scheduling and monitoring for the data pipeline using cron jobs and basic shell scripting, ensuring timely availability of data for downstream analysis and reporting.

Interactive Energy Projection Visualisation (MSc Project)

- Designed and built interactive **Power BI** dashboards to visualise energy modelling outputs and compare projection scenarios for various policy interventions.
- Developed complex **DAX** measures and data models to aggregate and present energy supply, demand, and carbon emission data effectively to non-technical stakeholders.
- Utilized **Power Query** for data extraction, transformation, and loading from multiple CSV and database sources, ensuring data consistency across the reporting solution.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)